

Chapter 3 Summary: Exploring Forces

1. What Is Force?

A force is a push or a pull that can make something start moving, stop moving, speed up or slow down, or change its direction or shape. The chapter opens with a Jenga tower activity, asking why the tower doesn't collapse when a block is removed and what determines how hard you need to push or pull a piece. Everyday actions such as opening a door, climbing stairs, writing, or sitting on a chair all involve force, and even the wind and the Earth itself are constantly creating force.

2. Force Has Direction

Describing a force completely requires two things: how strong the push or pull is, and which direction it acts in. A group pushing a heavy table from different sides shows that the table's movement depends on both the strength and direction of everyone's push. Examples used to illustrate this include Jenga blocks, bumper cars changing speed and direction on collision, rockets needing downward thrust to launch upward, and bicycle brakes acting opposite to the direction of motion to slow the bike down.

3. Measuring Things: Standard Units

Before standard units existed, people used body-based measurements like the cubit (elbow to fingertip), hand-span, and foot-span — but since everyone's body is a different size, these were inconsistent. This led to the creation of standard units that are the same for everyone, everywhere:

Quantity	Units	Measuring Tool	Example
Length	centimetres (cm), metres (m), kilometres (km)	Metre ruler, measuring tape	A pencil (~15 cm), a classroom (~8 m), home distance (~3 km)
Mass	grams (g), kilograms (kg)	Weighing balance	An eraser (~5 g), a person (~25–35 kg)
Capacity/Volume	millilitres (mL), litres (L)	Measuring cylinder	A water bottle (~500 mL or 0.5 L)
Time	seconds (s), minutes (min), hours (h)	Clock, watch	A heartbeat (<1 s), a class period (~30 min), a school day (~6 h)
Temperature	degree Celsius (°C)	Thermometer	Room temperature (~25°C), body temperature (~37°C)

4. Measuring Force: The Newton and the Spring Balance

Just as length, mass, time, and temperature each have standard units, force also needed one. The standard unit of force is the newton (N), named after Sir Isaac Newton. As a helpful reference point, a 100-gram object exerts roughly 1 newton of force due to gravity — lifting a school bag might take 30–50 N, an adult pushing a car might take several hundred N, and a rocket launching can involve millions of newtons.

Force is measured using a **spring balance**, a tool with a spring inside that stretches more as more force is applied; a scale on the side shows the reading in newtons. To use it, you hang the spring balance from a secure place and attach the object to the hook at the bottom, then read where the pointer stops.

5. Types of Forces

Non-Contact Forces (act without touching)

- **Gravitational force:** pulls objects toward the Earth (or any large object) even through air, with no touching required — this is why dropped objects always fall downward rather than floating away, and why the Moon orbits the Earth.
- **Magnetic force:** a force of attraction or repulsion between magnets and certain materials like iron and steel, working from a distance. Every magnet has a North and South pole. Real-world uses include refrigerator door magnets, MRI machines that see inside the body without surgery, and high-speed maglev trains that float above the track using magnetic repulsion.

Contact Forces (require touching)

- **Muscular force:** the force your muscles exert when you directly push or pull something, such as kicking a ball, opening a door, or pushing a cart.
- **Frictional force:** covered in detail below.

6. Friction

Friction is the force that opposes motion when two surfaces rub against each other. Even the smoothest-looking surfaces have tiny bumps and valleys (invisible without a microscope) that catch against each other when surfaces touch, slowing things down and creating heat.

Friction: A Friend or a Foe?

Friction can help or hinder us, depending on the situation.

- **When friction helps (friend):** it lets us walk without slipping (which is why wet floors are marked with warning signs), grip objects, and stop safely — shoe soles and climbing wall surfaces are intentionally roughened for this reason.
- **When friction is a problem (foe):** it can make drawers stick, cause rusty bicycle chains to become harder to pedal, and make heavy furniture exhausting to slide across the floor.

Ways to Increase Friction (for grip and control)

- Make surfaces rougher (e.g., sandpaper on wood)
- Add weight or press harder on the surfaces
- Use materials with more grip, like rubber
- Create grooves or tread patterns, like tyre treads that prevent skidding

Ways to Decrease Friction (for smoother movement)

- Make surfaces smoother (e.g., polished bowling lanes)
- Use lubricants such as oil, grease, or even water (e.g., engine oil, playground slides)
- Use wheels or rollers instead of dragging objects
- Reduce the contact area, like the thin blades of ice skates

Real-world examples: sand-filled "emergency ramps" on steep roads use high friction to stop runaway trucks safely, while the supersonic Concorde airplane was designed with a long, pointed nose and sleek body specifically to reduce air friction and fly faster.

7. Putting It All Together

Returning to the Jenga tower analogy: gravity pulls the blocks down, friction holds them in place even when some are missing, and the muscular force used to remove a block must be just right — strong enough to overcome friction, but gentle enough not to topple the tower.

8. Chapter Assessment (Milestone)

The chapter closes with a full review covering everything above:

- **Multiple choice:** questions on the nature of force, units of length, spring balance measurements, and identifying non-contact forces.
- **Fill in the blanks:** covering friction, standard units, the spring balance's unit, thermometers, and magnetic repulsion.
- **Match the columns:** matching situations (e.g., an apple falling from a tree, brakes stopping a bicycle, a magnet picking up pins, pushing a shopping cart) to the correct force type (gravitational, frictional, muscular, magnetic).
- **Understand and explain:** analyzing statements about friction and identifying forces acting on a stationary book on a table.
- **Think, observe, apply and solve:** listing real examples of helpful/unhelpful friction, and solving a scenario about moving heavy furniture by choosing ways to increase or decrease friction as needed.
- **Measure and record activity:** estimating and then measuring the weight (force) of three objects of different sizes in newtons using a spring balance.